Solution: Number Range

Problem Statement

Given an array of numbers sorted in ascending order, find the range of a given number ‘key’. The range of the ‘key’ will be the first and last position of the ‘key’ in the array.

Write a function to return the range of the ‘key’. If the ‘key’ is not present return [-1, -1].

Example 1:

```
Input: [4, 6, 6, 6, 9], key = 6
Output: [1, 3]
```

Example 2:

```
Input: [1, 3, 8, 10, 15], key = 10
Output: [3, 3]
```

Example 3:

```
Input: [1, 3, 8, 10, 15], key = 12
Output: [-1, -1]
```

Solution

The problem follows the **Binary Search** pattern. Since Binary Search helps us find a number in a sorted array efficiently, we can use a modified version of the Binary Search to find the first and the last position of a number.

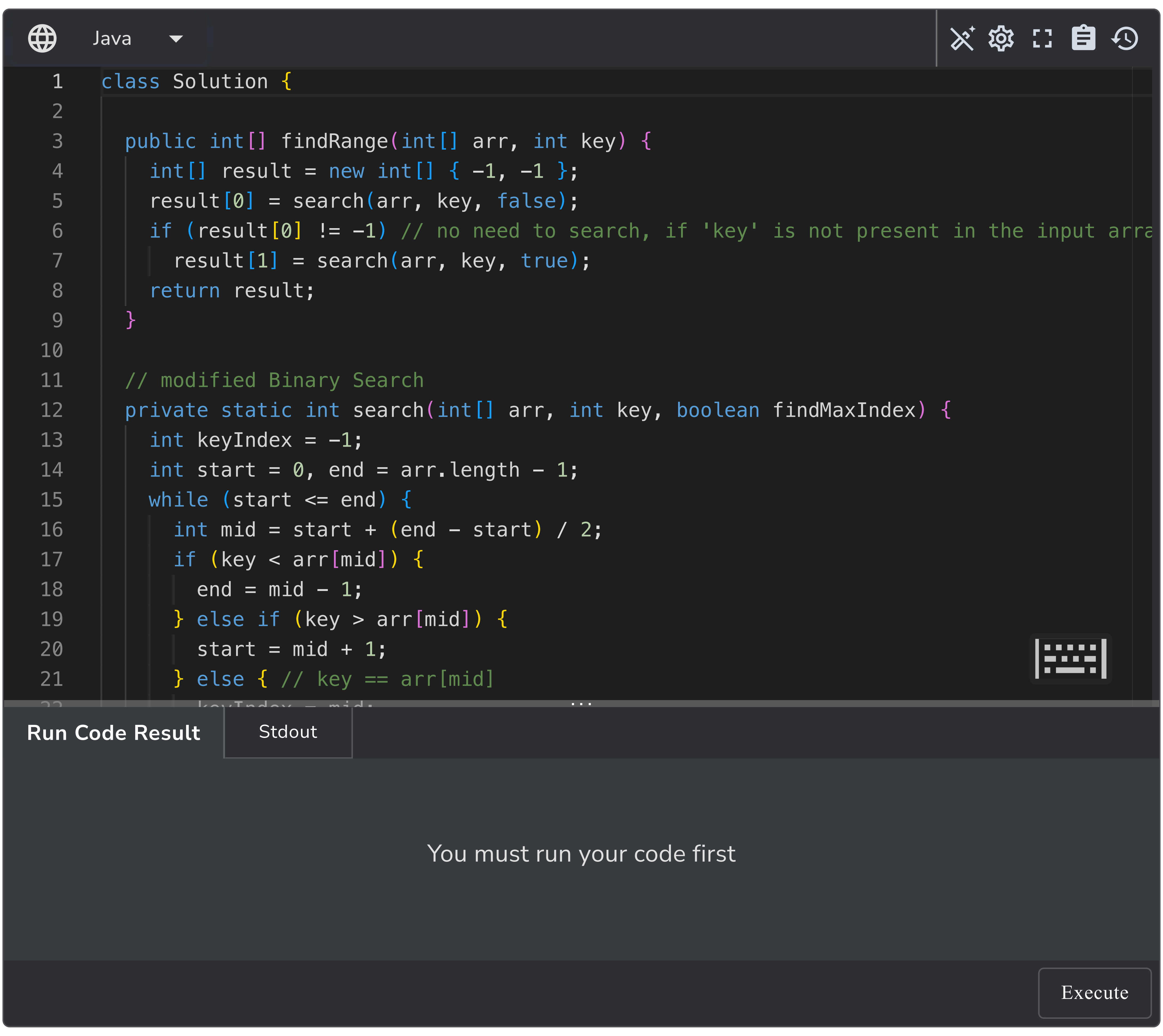
We can use a similar approach as discussed in `Order-agnostic Binary Search`. We will try to search for the ‘key’ in the given array; if the ‘key’ is found (i.e. `key == arr[middle]`) we have two options:

1. When trying to find the first position of the ‘key’, we can update `end = middle - 1` to see if the key is present before `middle`.
2. When trying to find the last position of the ‘key’, we can update `start = middle + 1` to see if the key is present after `middle`.

In both cases, we will keep track of the last position where we found the ‘key’. These positions will be the required range.

Code

Here is what our algorithm will look like:



```java
class Solution {

  public int[] findRange(int[] arr, int key) {
    int[] result = new int[] { -1, -1 };
    result[0] = search(arr, key, false);
    if (result[0] != -1) // no need to search, if 'key' is not present in the input arra
      result[1] = search(arr, key, true);
    return result;
  }

  // modified Binary Search
  private static int search(int[] arr, int key, boolean findMaxIndex) {
    int keyIndex = -1;
    int start = 0, end = arr.length - 1;
    while (start <= end) {
      int mid = start + (end - start) / 2;
      if (key < arr[mid]) {
        end = mid - 1;
      } else if (key > arr[mid]) {
        start = mid + 1;
      } else { // key == arr[mid]
```

You must run your code first

Time Complexity

Since, we are reducing the search range by half at every step, this means that the time complexity of our algorithm will be O(logN) where ‘N’ is the total elements in the given array.

Space Complexity

The algorithm runs in constant space O(1).